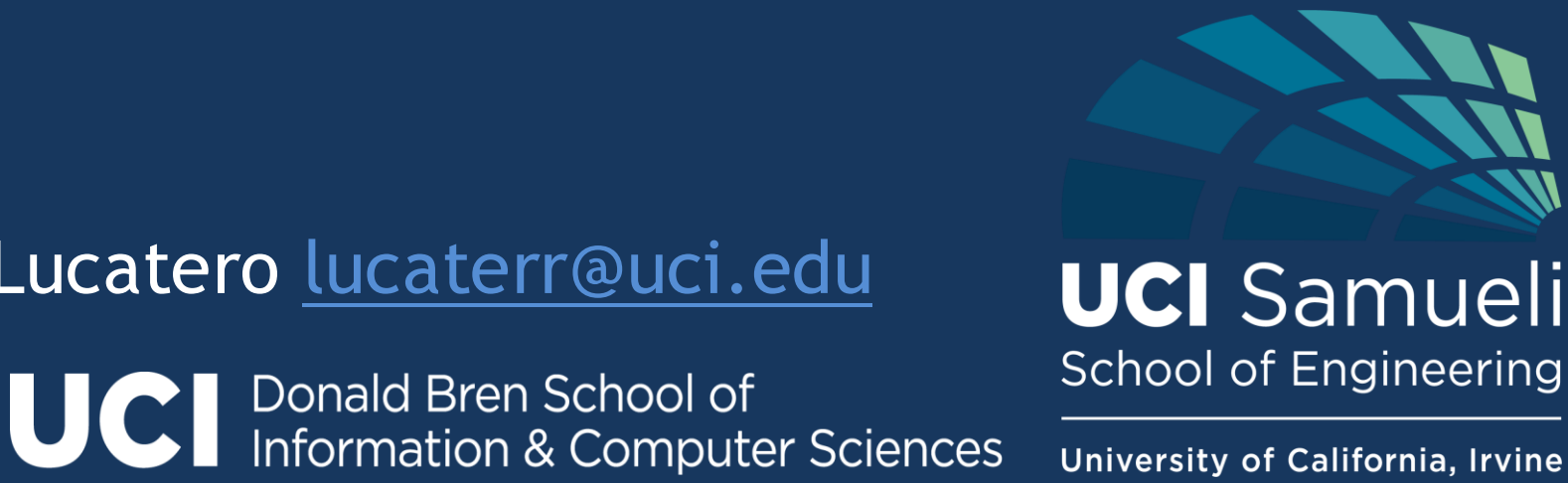




3D-Printed AI-Powered Microscope

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Abstract

We developed a low-cost, 3D-printed motorized microscope based on the OpenFlexure High-Resolution model. During Phase 1, we fully assembled the hardware, integrated the Raspberry Pi, Sangaboard, and stepper motors, and verified functionality through the OpenFlexure Web Application Interface. The system successfully captured clear images of biological samples and demonstrated stable automated focusing and stage motion. These results establish a reliable foundation for Phase 2, where AI-assisted cell counting/analysis and a custom user interface will be implemented.

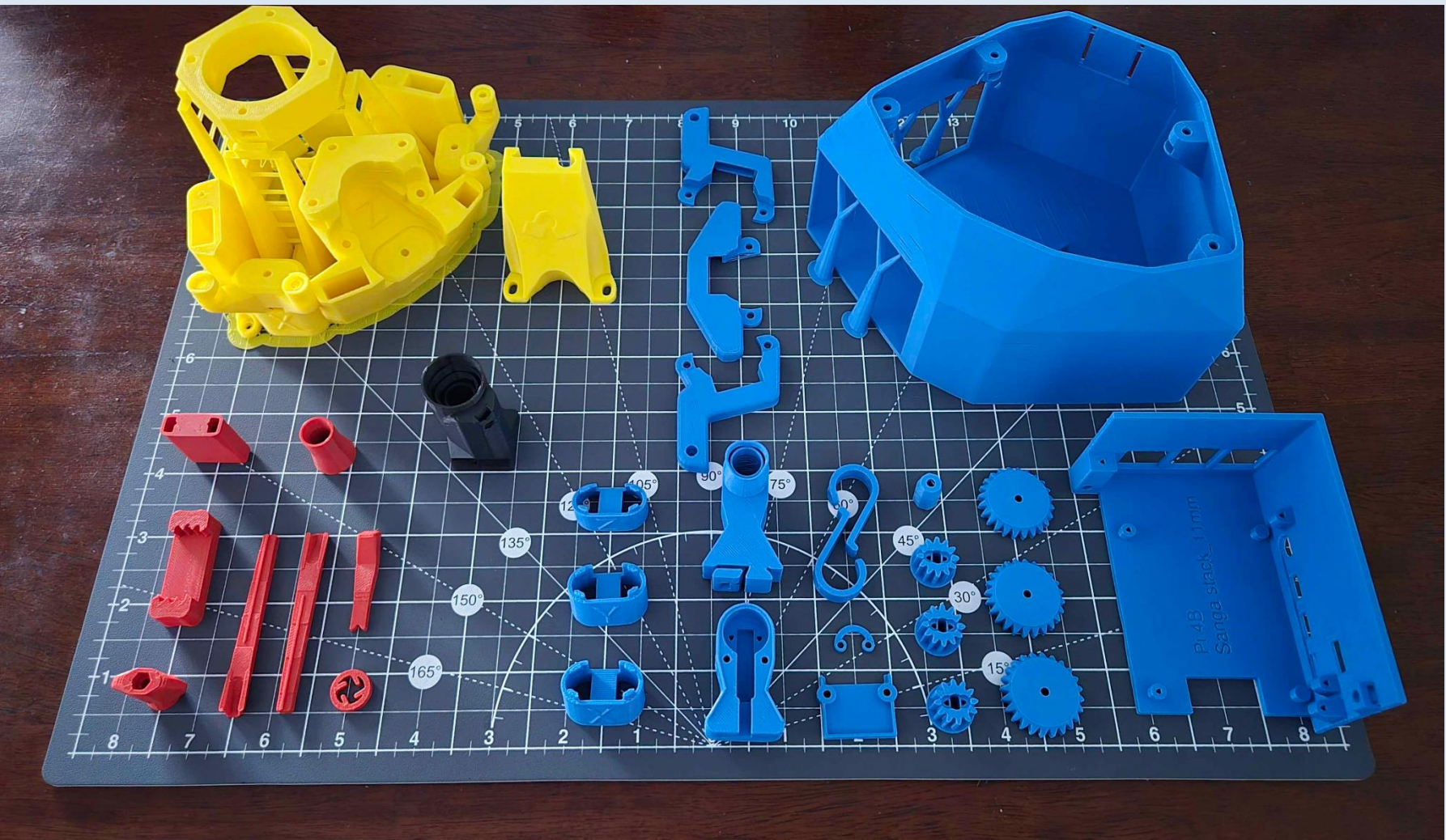
Objectives

- Develop a 3D-printed microscope based on the OpenFlexure High-Resolution Motorized model.
- Ensure full hardware and software functionality, including compatibility with the OpenFlexure Web Application Interface.
- Establish the foundation for Phase 2, where automated cell counting will be integrated through machine learning, computer vision, and a custom user interface.
- Provide a low-cost, sustainable, and accessible imaging tool for students, educators, and research professionals across diverse environments.

Materials and Methods

- We purchased a kit that contained our starting hardware
- Raspberry Pi 4 Model B
 - Raspberry Pi Camera Board v2 w/ 20cm long flex ribbon cable
 - Sangaboard v0.5
 - 3 x 28BYJ-48 5V stepper motors
 - 64GB High Speed MicroSD Card
 - Condenser Lens (5mm focal length plastic lens)
 - Illumination LED board w/ 12" long 2-pin cable & diffusers
 - 12.7mm achromatic doublet lens
 - 3D printed microscope shell
 - 40X Objective Microscope Lens

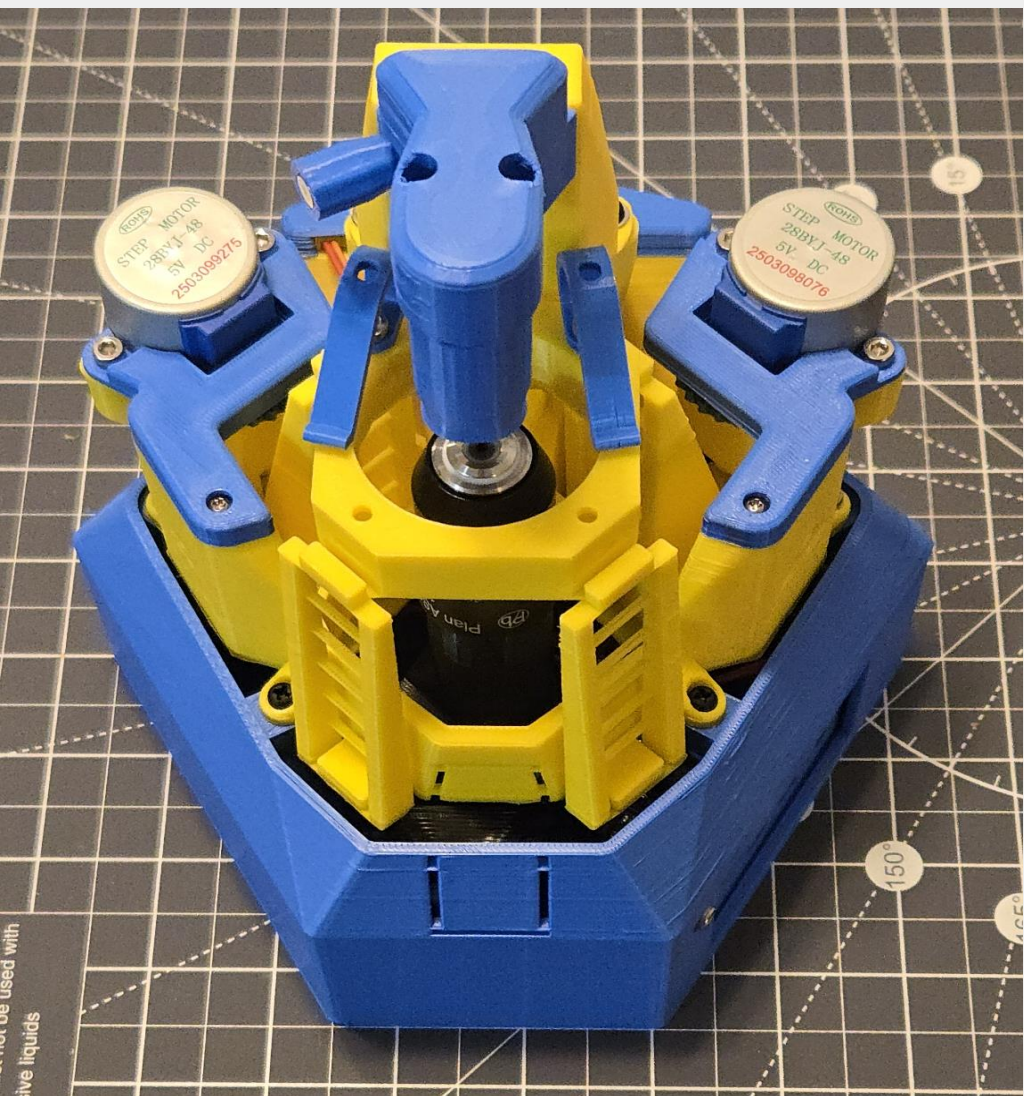
Diagrams/Figures/Experiments



3D-Printed Components

All PLA components were 3D-printed in a functional color scheme: red parts are assembly tools not included in the final microscope; the black optical adapter reduces reflections and sets the correct spacing for the 40× objective; yellow pieces form the main body and illumination housing; and blue components make up the base and remaining structural supports.

The stepper motors, gear system, and lens holder were assembled and aligned to enable automated XYZ movement. The assembled unit integrates the Raspberry Pi, camera module, illumination system, and motorized motion platform.



Mechanical Assembly



Test image of red onion cells.

Captured using the OpenFlexure Web Interface. The microscope successfully imaged cell walls and demonstrated stable focusing and illumination.

Results

We achieved our Phase 1 goal of building and testing a functional OpenFlexure *High-Resolution Motorized* model, which is the standard research configuration, and the foundation for Phase 2 of the project.

In addition to confirming its functionality with hardware and software diagnostics, we also captured a test image of red onion cells.

Additional Information

By 3D-printing our own components, we reduced overall costs compared to purchasing a full OpenFlexure accessories kit. The hardware-only kit costs approximately \$135, while the version including 3D-printed parts costs about \$200, meaning our approach saved roughly \$65 in printed-component expenses. Although we purchased the 40× objective lens separately for about \$55, our total Phase 1 cost remained within the expected budget range for assembling the OpenFlexure High-Resolution Motorized configuration. Planned Phase 2 upgrades include adding the Google Coral USB accelerator for faster machine-learning inference, along with improvements to image handling, quality, and overall system performance.

References

- [1] OpenFlexure High-Resolution Motorized Configuration. <https://build.openflexure.org/openflexure-microscope/v7.0.0-beta4/#high-resolution-motorised-microscope>
- [2] OpenFlexure Web Application Interface. <https://openflexure-microscope-software.readthedocs.io/en/latest/webapp/index.html>
- [3] OpenFlexure Microscope – Source Files and Design Repository. <https://gitlab.com/openflexure/openflexure-microscope>